Alexander Oh

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Education

National University of Singapore, Singapore, SG — Aug 2023 – Jan 2027 (Expected)
Bachelor of Science in Computer Science

Professional Experience

Tetrix (Fintech Startup), New York, NY — Jan 2026 – Present
Backend Software Engineer

- Built a data pipeline for extracting and transforming financial data from 50K+ documents; parallelized feature extraction to increase processing throughput 30x while achieving ~95% accuracy and ~100% precision for a coordinate tracing model.
- Spearheaded a scalable data extensibility framework and custom field system across 6 microservices, supporting 15+ private-market data models and reducing schema-change overhead by ~80%.
- Optimized large-scale financial computations with NumPy vectorization and a DAG-based calculation engine with topological dependency resolution, improving time-series processing performance by 5–10x on 100K+ row datasets.
- Designed a distributed FX conversion service using Redis Pub/Sub for cross-pod cache invalidation and updates, efficiently refreshing 10 years of historical FX data and achieving 4x faster performance of a Postgres-backed implementation.
- Engineered a 4-stage SQS pipeline with fan-out orchestration and content-based hashing for deduplication; implemented SQL row-level locking for concurrency control, generating 500+ Portfolio Variance Reports with 0 duplicate jobs.

TikTok, Singapore, SG — June 2025 – Nov 2025
Backend Software Engineer for User Growth (Paid Ads)

- Implemented a robust state-machine framework to manage complex payment approval workflows, cleanly decomposing 100+ conditional execution branches to minimize system edge cases.
- Eliminated payment duplication errors in distributed workflows by implementing async SQL locking mechanisms, achieving 0 duplicate approval incidents across 100+ peak transactions/hour.
- Implemented a high-throughput batch upload system with integrated deduplication and validation algorithms, leveraging asynchronous cron jobs to process 1,000+ concurrent files without impacting user-facing performance.

Asian Institute of Digital Finance (Fintech Research Institute), Singapore, SG — Jan 2025 – May 2025
Undergraduate Research Intern

- Built Python/Pandas pipelines to reconcile legacy database variables during a Bloomberg migration using OpenAI APIs and fuzzy matching, reducing estimated manual migration effort from ~10 weeks to under 5 days
- Refactored the Python backend to integrate new databases, rewriting outdated queries while implementing comprehensive error handling and multilingual input support.

Oneberry Technologies (Security Enterprise), Singapore, SG — May 2024 – Aug 2024
Fullstack Software Engineer

- Built a real-time monitoring platform capable of concurrently processing and streaming telemetry from 10,000+ live IoT sensors with low-latency updates (60k+ per hour).
- Led backend development to build scalable event-driven messaging and websocket communication infrastructure.
- Deployed and maintained cloud-hosted services on DigitalOcean using PM2 process orchestration to manage concurrent workers, improving system reliability and supporting live production-scale testing.

Projects

Personal Website — Dec 2025 – Present
https://github.com/alexandtheoh/alexander_oh_website

- Developed a browser-native context aware RAG pipeline using WebLLM and Gemma-2 2B, leveraging TypeScript-implemented cosine similarity and top-k retrieval over 10K+ pre-processed indexed chunks.
- Engineered a conversation compression layer to dynamically prune and summarize chat state, maximizing token efficiency within strict client-side hardware limits.

Whatsapp LLM — May 2025 – Present
https://github.com/alexandtheoh/whatsapp_llm

- Fine-tuned a Mistral 7B language model on 250K+ conversational messages using QLoRA to explore personalized dialogue generation and conversational style adaptation.
- Designed custom conversational training templates and preprocessing pipelines to preserve multi-turn dialogue structure and long-range conversational context.

Skills and Interests

- **Languages:** Python, C++, Go, Java, TypeScript, SQL, VBA
- **Libraries & Frameworks:** NumPy, Pandas, scikit-learn, Node.js, FastAPI, React, Thrift/Kitex, Hertz
- **Systems & Infrastructure:** Docker, RabbitMQ, MQTT, Microservices, Socket.io
- **Interests:** Tennis & Track (Represented NUS Club), Basketball, Golf, Hiking, Pokemon Showdown